

Controller Tests 01.12.25

Name	KI	r1	r2	q	Observer / Controller Speed	Phase Margin	Gain Crossover freq
Controller 1	6	8	4	10^{-3}	6.56	77	1.46
Comment	safe controller, fast observer						
Controller 2	15	20	1	10^{-2}	3.52	64	2.05
Comment	smal Phase margin, Large KI reletively high r in comparison, relativley slow observer						
Controller 3	6.5	3	0.5	10^{-3}	5.62	63.4	2.37
Comment	smal Phase margin, smal KI, smal r, faster observer						
Controller 4	5.5	1	0.1	$0.5 \cdot 10^{-3}$	4.28	63.1	3.25
Comment	smal Phase margin, smal KI, very smal r, fast observer						
Controller 5	8	1	0.06	$0.5 \cdot 10^{-3}$	4.14	62.3	3.9645
Comment	borderline unsafe, smal overshoot						